

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-KC-4501-04 Sheet 1 of 1
OPL Title	Anti-Surge / Recycle Operation	Discipline	Process / Operations	
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	Area / Unit	4500 - RECYCLE GAS SYSTEM	
Related Interlock	SEQ-4501 (SIL 2)	P&ID Ref	TJC-LLD-PID-4501	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **anti-surge / recycle operation** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. The recycle (spillback) valve protects the machine at low throughput and during start/stop; incorrect operation risks deadhead and rod overload. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Keep the anti-surge FV-4502 in auto during start-up	Permit valid & briefed
2	Avoid deadheading against a high discharge pressure (PSHH-4502)	Parameter within limit
3	Watch suction PSL-4501 for gas starvation	Interlock status confirmed
4	Return the machine to load slowly, closing recycle gradually	Change logged on DCS
5	Investigate any repeated surge cycling with the vendor curve	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- The recycle (spillback) valve protects the machine at low throughput and during start/stop; incorrect operation risks deadhead and rod overload.
- Respect the interlock SEQ-4501: e.g. PSL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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